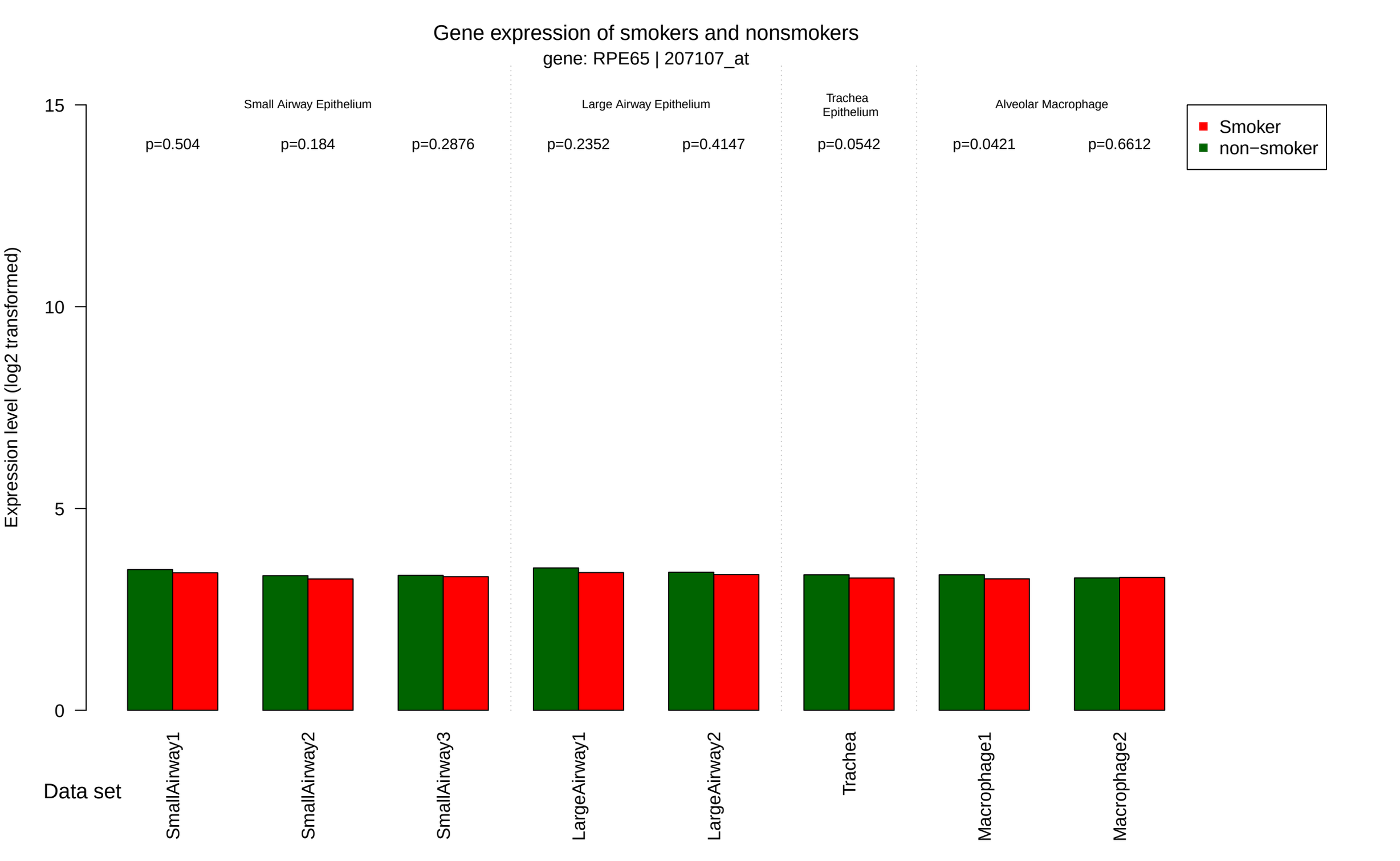
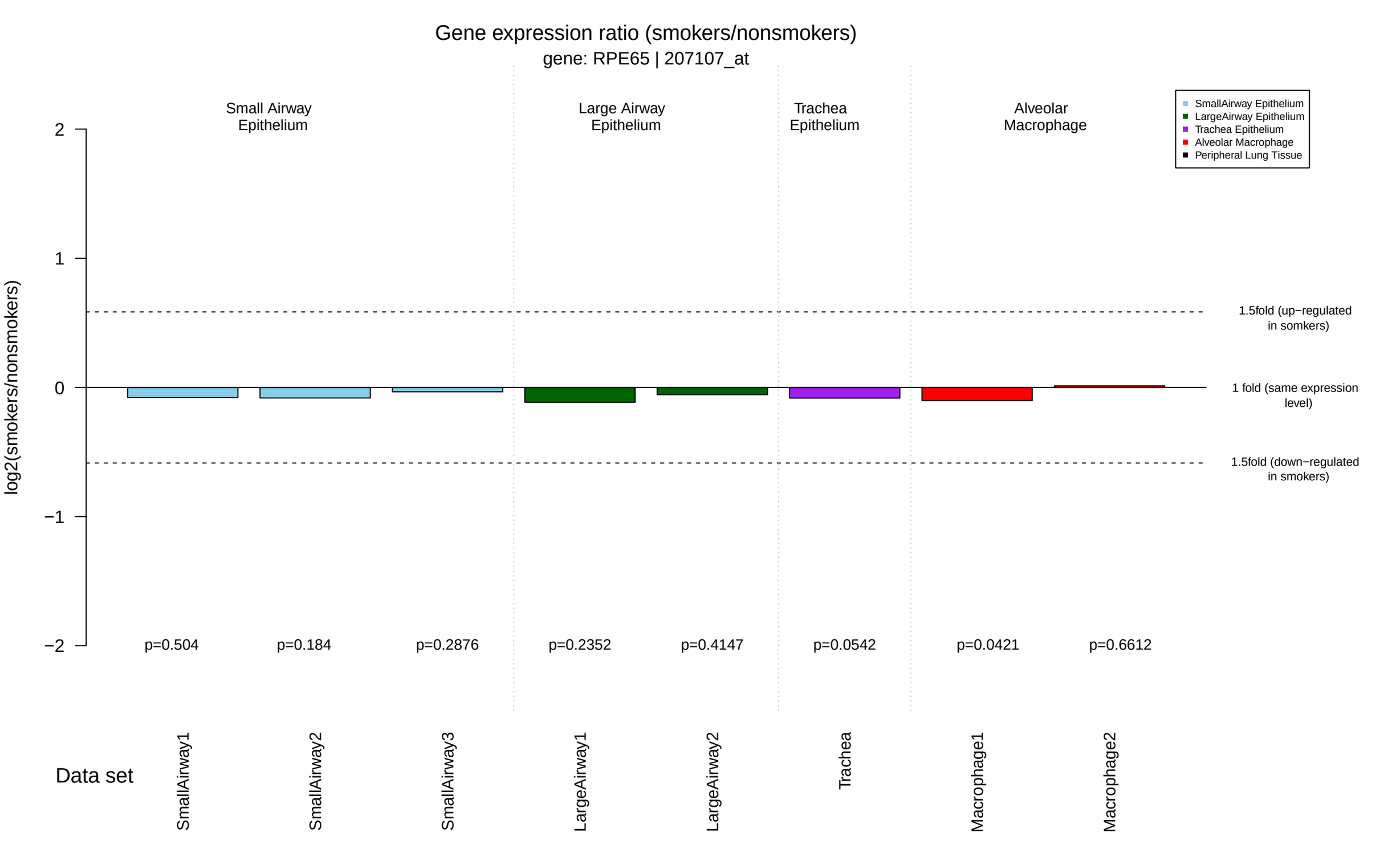
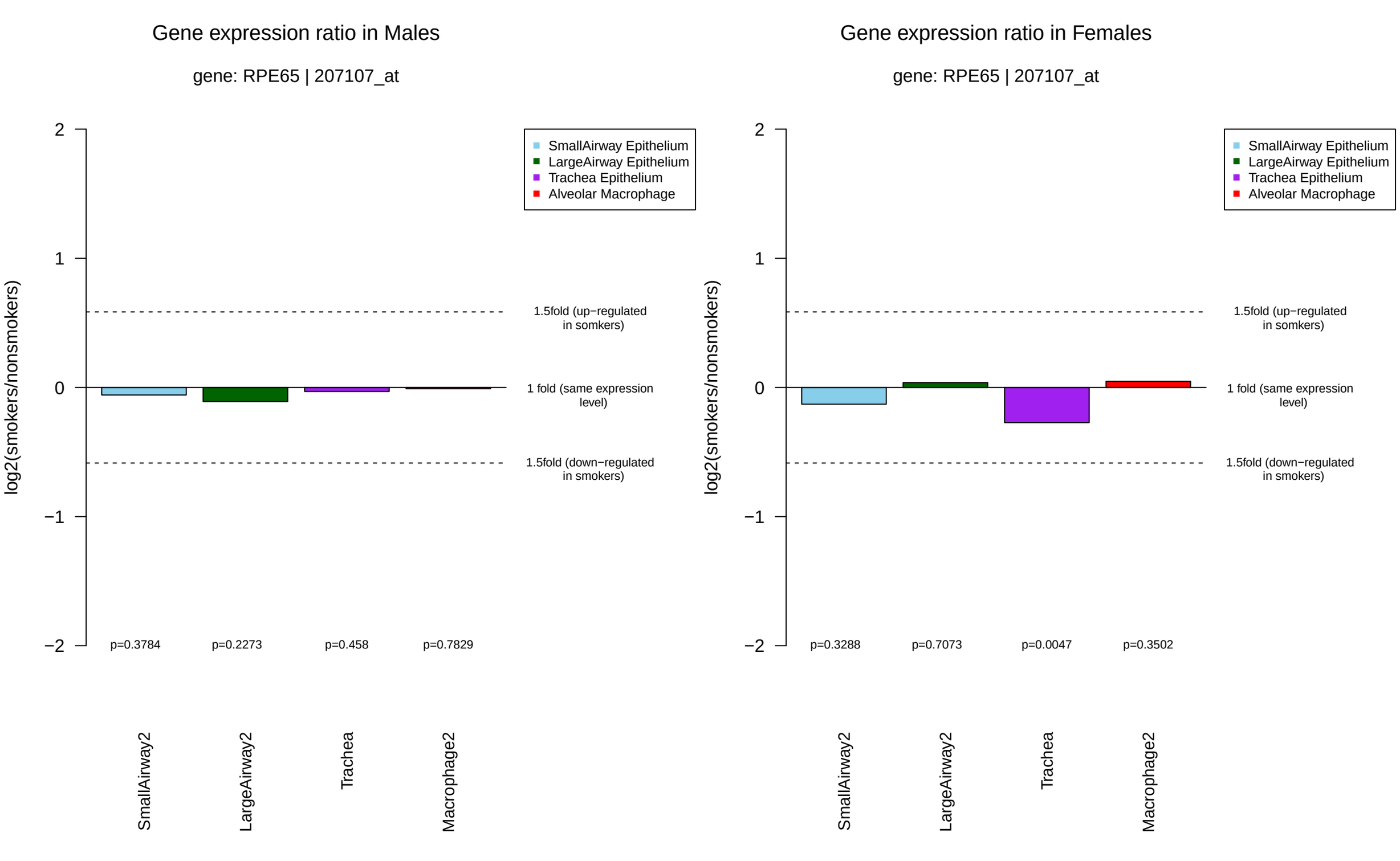
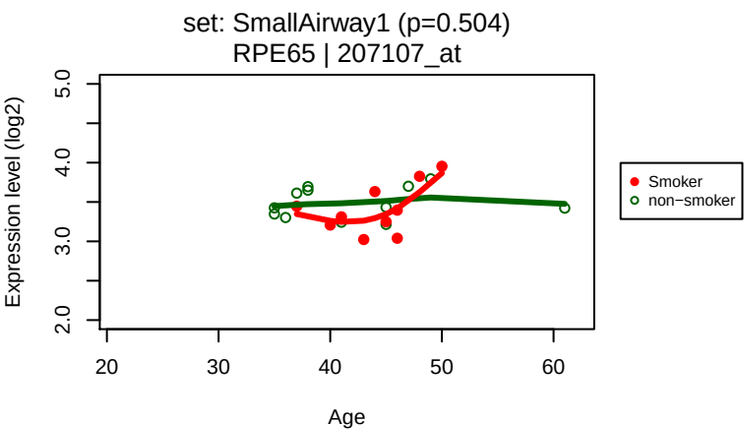
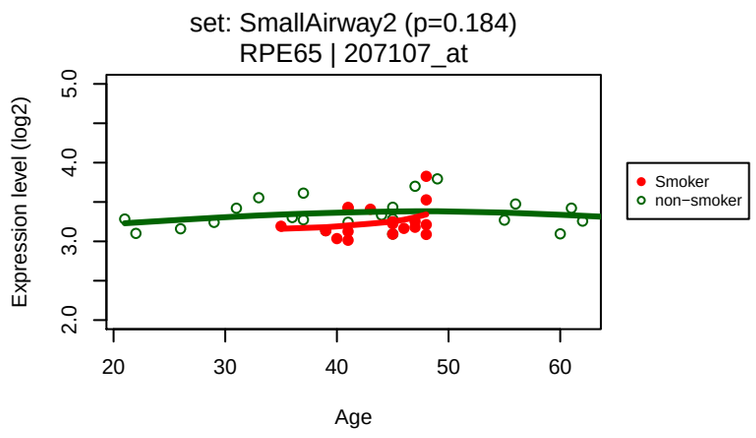
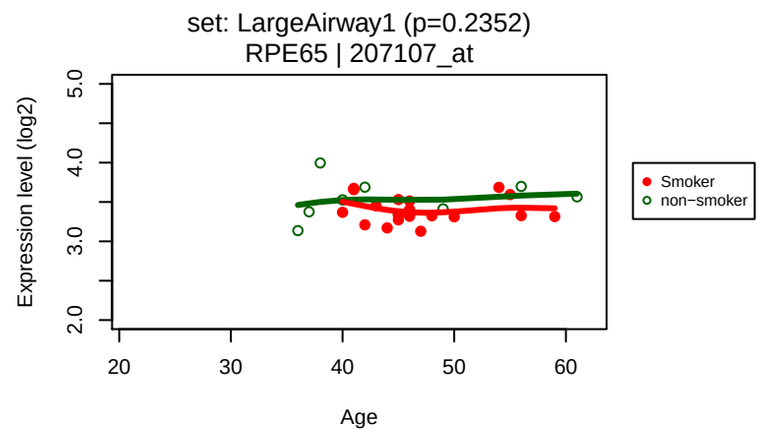

Gene expression of smokers and nonsmokers
gene: RPE65 | 207107_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.504
p=0.184
p=0.2876
p=0.2352
p=0.4147
p=0.0542
p=0.0421
p=0.6612
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

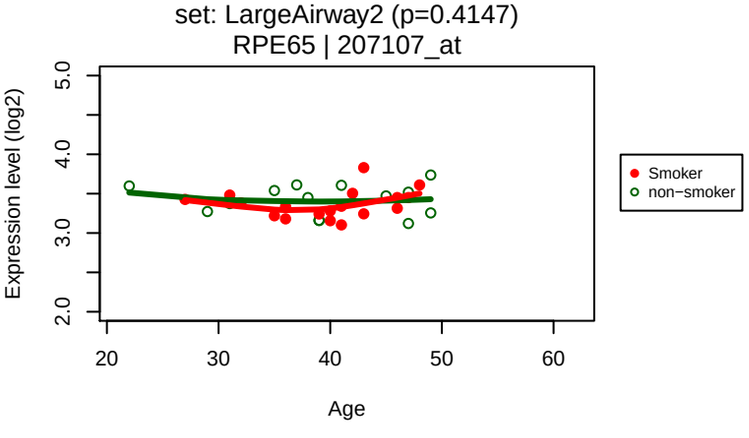

Gene expression ratio (smokers/nonsmokers)
gene: RPE65 | 207107_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.504
p=0.184
p=0.2876
p=0.2352
p=0.4147
p=0.0542
p=0.0421
p=0.6612
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

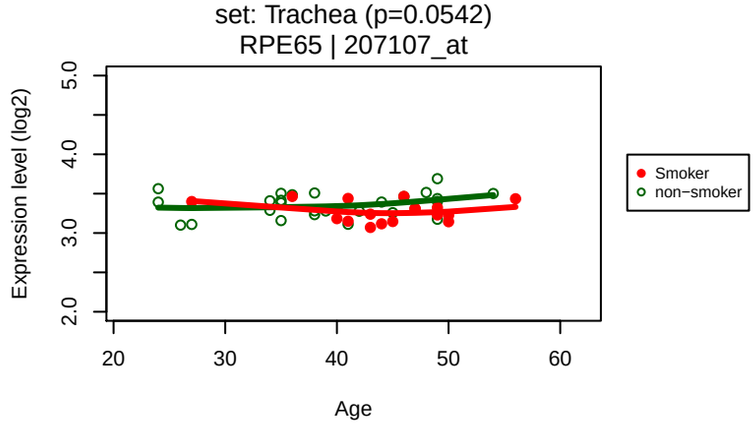

Gene expression ratio in Males
gene: RPE65 | 207107_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3784
p=0.2273
p=0.458
p=0.7829
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPE65 | 207107_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3288
p=0.7073
p=0.0047
p=0.3502
SmallAirway2
LargeAirway2
Trachea
Macrophage2

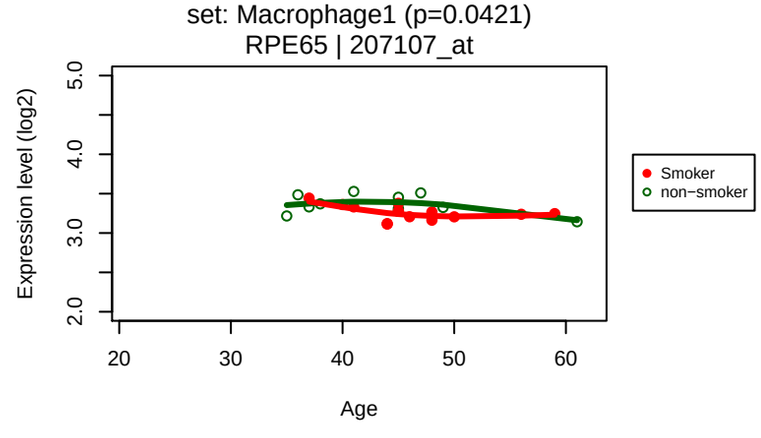

set: SmallAirway1 (p=0.504)
RPE65 | 207107_at
Expression level (log2)
Age
Smoker
non−smoker

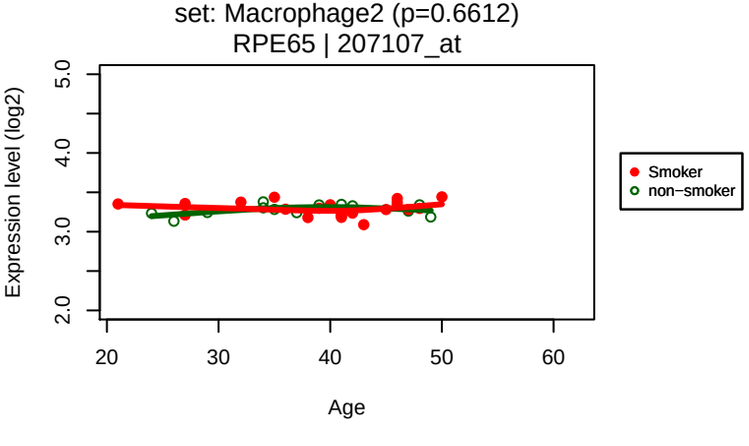

set: SmallAirway2 (p=0.184)
RPE65 | 207107_at
Expression level (log2)
Age
Smoker
non−smoker

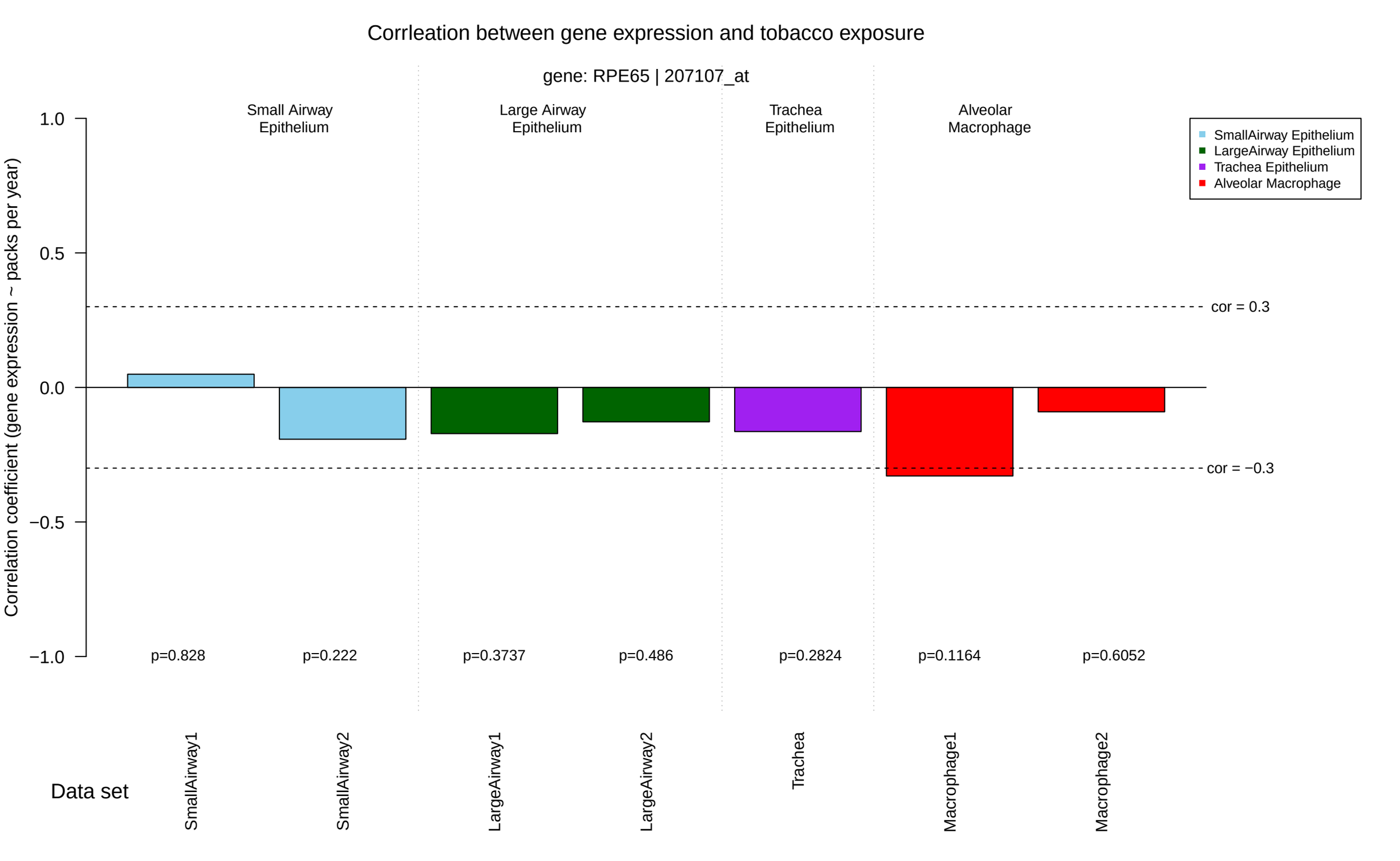

set: LargeAirway1 (p=0.2352)
RPE65 | 207107_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4147)
RPE65 | 207107_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0542)
RPE65 | 207107_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0421)
RPE65 | 207107_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6612)
RPE65 | 207107_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPE65 | 207107_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.828
p=0.222
p=0.3737
p=0.486
p=0.2824
p=0.1164
p=0.6052
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2